

Function notation



- 1 The point $(7, -6)$ satisfies the function $f(x)$. Write this using function notation.
- 2 If $f(x) = 9x^2 + 7x - 4$, find:
 - a $f(-4)$
 - b $f(10)$
- 3 If $f(x) = -6x + 4$, find:
 - a $f(4)$
 - b $f(0)$
- 4 If $f(x) = 4x + 4$, find:
 - a $f(2)$
 - b $f(-5)$
- 5 If $f(x) = 3x - 1$, find:
 - a $f(3)$
 - b $f(-4)$
- 6 If $g(x) = \frac{7x}{4}$, find:
 - a $g(5)$
 - b $g(-4)$
- 7 Consider the function $f(x) = 4 + x^3$.
 - a Evaluate $f(4)$
 - b Evaluate $f(-2)$
- 8 Consider the function $f(x) = 2x^2 - 2x + 5$. Evaluate $f\left(\frac{1}{2}\right)$.
- 9 Consider the function $f(x) = \sqrt{5x + 9}$. Find the exact value of:
 - a $f(0)$
 - b $f(2)$
 - c $f(-1)$
- 10 If $f(t) = \frac{t^3 + 27}{t^2 + 9}$, evaluate the following:
 - a $f(-3)$
 - b $f(3)$
 - c $f(4)$
- 11 Consider the function $f(x) = x^2 + 8x$. Write an expression for:



12 Consider the function $f(x) = 2x^3 + 3x^2 - 4$.

a Evaluate $f(0)$.

b Evaluate $f\left(\frac{1}{4}\right)$.

13 If $j(x) = 3^x - 3^{-x}$, find the following, rounding your answers to two decimal places if necessary:

a $j(0)$

b $j(1)$

c $j(4)$

14 If $m(x) = \sqrt{12^2 - x^2}$, find:

a $m(4)$

b $m(0)$

c $m(9)$

d $m(\sqrt{2})$

15 Consider the function $p(x) = x^2 + 8$.

a Evaluate $p(2)$.

b Form an expression for $p(m)$.



16 Consider the equation $x + 3y = 6$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f(x)$ for y .

c Find the value of $f(3)$.



17 Consider the equation $x - 4y = 8$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f(x)$.

c Find the value of $f(12)$.



18 Consider the equation $y + 6x^2 = 3 - x$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f(x)$.

c Find the value of $f(3)$.



19 Consider the equation $y - 4x^2 = 5 + x$.

a Make y the subject of the equation.

b Rewrite the equation using function notation $f(x)$.

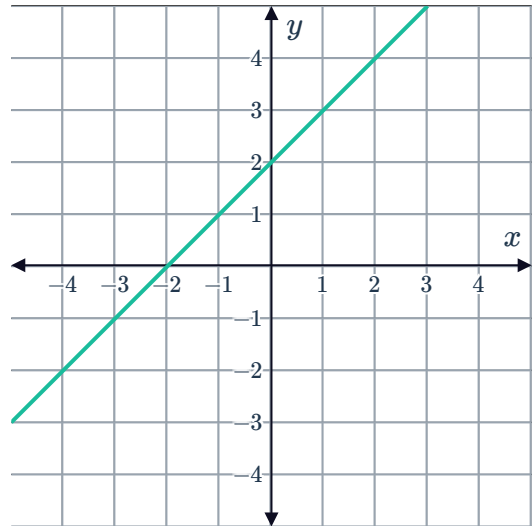


20 Consider the equation $-6x + 5y = 7$.

- a Make y the subject of the equation.
- b Rewrite the equation using function notation $f(x)$.
- c Find the value of $f(3)$.

21 Use the graph of the function $f(x)$ to find each of the following values:

- a $f(0)$
- b $f(-2)$
- c The value of x such that $f(x) = 3$



22 Consider the relation $h(x) = -x^2 + 6x - 6$.

- a For any input value of x , state the maximum number of distinct output values $h(x)$ can produce.
- b Is $h(x)$ a function? Explain your answer.

23 For each of the following pairs of variables, determine which is the independent variable and which is the dependent variable:

- a Cost of pizza and size of pizza.
- b Mark achieved on a test and time spent studying.
- c Duration of a loan and amount borrowed.
- d Time spent exercising and physical fitness level.

24 Consider the function $f(x) = x^2 - 49$.

- a Find:
 - i $f(1)$
 - ii $f(8)$
 - iii $f(0)$
- b If $f(x) = 12$, what are the possible values for x , rounded to two decimal places?

25 Consider the function $g(x) = ax^3 - 3x + 8$



- a Form an expression for $g(k)$.
- b Form an expression for $g(-k)$.
- c Is $g(k) = g(-k)$?
- d Is $g(k) = -g(-k)$?

Algebra of functions

26 A function is defined as $f(x) = 2 + 3x - x^3$. Evaluate $f(-2) + f(4)$.

 **27** If $A(x) = x^2 + 1$ and $Q(x) = x^2 + 9x$, evaluate the following:

- a $A(5)$
- b $Q(4)$
- c $A(3) + Q(2)$

 **28** Consider the function $f(x) = x^3 - 8x + 7$.

- a Evaluate $f(4)$.
- b Evaluate $f(-4)$.
- c Form an expression for $f(a)$.
- d Form an expression for $f(b)$.
- e Form an expression for $f(a + b)$.
- f Is $f(a) + f(b) = f(a + b)$ for all values of a and b ?

29 Consider the function $f(x) = x^2 + 5x$. Find $f(a + h)$.

30 Consider the function $f(x) = x^2 - 2x$. Find $f(a + 6)$.

31 Consider the function $f(x) = x^2 - 3x - 2$.

- a Form an expression for $f(a - 2)$.
- b Form an expression for $f(a + h) - f(a)$.

32 Consider the function $f(x) = 4x^2 + 4x + 3$.

- a Form an expression for $f(a + 2)$.
- b Form an expression for $f(a + h) - f(a)$.

- 33 Consider the following table of values:



x	2	7	8	9
$f(x)$	4	14	16	18
$g(x)$	8	28	32	36

- a Find $(f + g)(2)$. b Find $(f + g)(7)$.
c Find $(f \times g)(2)$. d Find $(g - f)(9)$.

- 34 Let $f(x) = x^2 - 1$ and $g(x) = 5x - 1$. Find $f(2) + g(2)$.

- 35 Let $f(x) = -5x + 3$ and $g(x) = x^2 - 7$.

- a Find $(f \times g)(x)$. b Evaluate $(f \times g)(-3)$.

- 36 If $f(x) = 3x - 5$ and $g(x) = 5x + 7$, find:

- a $(f + g)(x)$ b $(f + g)(4)$ c $(f - g)(x)$ d $(f - g)(10)$

Applications

- 37 If $Z(y) = y^2 + 12y + 32$, find y when $Z(y) = -3$.

- 38 A function $f(x)$ is defined by $f(x) = (x + 4)(x^2 - 4)$.

- a Evaluate $f(6)$. b Find all solutions for which $f(x) = 0$.

- 39 A graph of a quadratic equation of the form $y = ax^2 + bx + c$ passes through the points $(0, -7)$, $(-1, -8)$ and $(4, 77)$.

- a Using the point $(0, -7)$, find the value of c .
b Substitute c and $(-1, -8)$ into the equation $y = ax^2 + bx + c$ to obtain an equation that describes a in terms of b .
c Substitute c and $(4, 77)$ into the equation $y = ax^2 + bx + c$ to obtain a simplified second equation in terms of a and b .
d Solve for a and b .

40 The financial team at The Gamgee Coopers  wants to calculate the profit, $P(x)$, generated by producing x units of wetsuits.

The revenue produced by the product is given by the equation is $R(x) = -\frac{x^2}{4} + 40x$. The cost of production is given by the equation $C(x) = 5x + 410$. The profit is calculated as $P(x) = R(x) - C(x)$.

- a** Find an expression for $P(x)$ in terms of x .
- b** Find the values of the following:
 - i** $R(70)$
 - ii** $C(70)$
 - iii** $P(70)$
- c** Sketch the graphs of $y = R(x)$, $y = C(x)$ and $y = P(x)$.

41 Elizabeth wants to calculate the cost to travel to Tehran and then Mexico City at certain times of the year. A program on her computer calculates $T(x)$, the total cost of this trip, by adding the cost of travelling to Tehran, $S(x)$, and the cost of travelling to Mexico City, $U(x)$. Based on historical data, the computer program uses the calculations:

$$S(x) = x^2 - 200x + 10\,227 \quad \text{and} \quad U(x) = 18x + 15\,423$$

where x is the number of days until the date of travel.

- a** Find an equation for $T(x)$.
- b** Find the values of the following:
 - i** $S(91)$
 - ii** $U(91)$
 - iii** $T(91)$
- c** Sketch the graphs of $y = S(x)$, $y = U(x)$ and $y = T(x)$.